

YAN-CHENG (BILL) HSU

Infrastructure Software Engineer | AI Platform & Kubernetes Expert

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PROFESSIONAL EXPERIENCE

Alibaba Cloud U.S. | Sunnyvale, CA

Infrastructure Software Engineer II | AnalyticDB Org, AI Training Platform | Jul 2025 – Present

Lead Engineer for Next-Gen Cross-Cluster AI PaaS: A serverless training orchestration platform for large-scale distributed workloads.

- **Architected "K8s-on-K8s" Recursive Virtualization:** Designed a dual-layer Virtual Kubelet system to decouple Control Plane (CPU-rich) from Data Plane (GPU-rich). Abstracted a massive-scale heterogeneous fleet into region-wide, SKU-partitioned resource pools, enabling 25x capacity scaling and seamless Serverless-to-Reserved migration (achieving ~40% TCO reduction).
- **Engineered Cross-Cluster Identity Mesh:** Solved critical "split-brain identity" in distributed systems by building a Credential Projection Controller. Implemented automated SA token rotation and secure secret injection, enabling remote Ray Workers to authenticate with Central Master API without exposing static credentials.
- **Solved Network Segmentation for Distributed AI:** Overcame VPC overlay network isolation by designing a Network Piercing Injector. Bypassed cross-cluster DNS failures by dynamically rewriting Pod specs to map internal service endpoints to Internal CLBs (L4), ensuring low-latency connectivity for Ray GCS and Autoscalers across physical boundaries.
- **Built Automated Telemetry & Reconciliation:** Developed custom Kubernetes Controllers to reconcile remote GPU node health. Integrated hardware metrics (DCGM) and eBPF kernel traces into a centralized telemetry pipeline, drastically reducing mean-time-to-detect (MTTD) for "zombie workers".

Amazon (AWS) | Seattle, WA

Software Development Engineer | High Performance Computing (HPC) & AI | Oct 2023 – Jul 2025

Core contributor to Massive-Scale GPU Infrastructure: The backbone for training foundation models.

- **Orchestrated GPU Lifecycle Management:** Architected a scalable tracking system handling high-concurrency scaling events during peak training cycles. Designed a parent-child DAG in Apache Airflow to eliminate circular termination loops, delivering multi-million dollar annualized savings via optimized resource reclamation.
- **Automated Fault Remediation:** Engineered a "Divide-and-Conquer" fault identification workflow for large-scale GPU clusters. Utilized dynamic K8s node labeling to isolate and diagnose hardware failures, reducing engineering troubleshooting time by approx. 90% per incident.
- **Optimized API Performance:** Re-architected customer-facing CLI/API (Lambda, Smithy) and optimized DynamoDB access patterns ($O(N) \rightarrow O(\log N)$), improving query response latency by over 70%.

TECHNICAL SKILLS

- **Core:** Distributed Systems, System Design, Cloud-Native Architecture, High Availability
- **Languages:** Golang (Expert), Python, C/C++, Java, TypeScript, SQL
- **AI Infrastructure:** Kubernetes (Internals, Operators, Virtual Kubelet, CNI/CSI), Ray (Cluster/Serve), HPC, PyTorch Distributed
- **Cloud & Tools:** AWS (EKS, Lambda, DynamoDB), Alibaba Cloud (ACK, ECI), Terraform, Prometheus, Grafana, eBPF

EDUCATION

M.S., Computer Science (Artificial Intelligence) | UC San Diego

2021 – 2023

B.S., Electrical Engineering | National Central University, Taiwan

2015 – 2019

PUBLICATIONS

- Amazon AGI et al. "The Amazon Nova Family of Models: Technical Report." 2024
- Hsu, Y.C. et al. "Generalized DNN Model for Cuffless Blood Pressure Estimation." Sensors, 2020
- Latifa, N.; Hsu, Y.C. et al. "Optimal Self-Supervised Multi-Fault Detector for Temperature Sensor Data." APSIPA, 2023